2024 6th International Conference on Blockchain Computing and Applications (BCCA) | 979-8-3503-5153-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/BCCA62388.2024.10844460

Optimal Portfolios Of Liquidity Positions

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***Abstract*—This paper introduces a novel quantitative framework for optimizing liquidity provider (LP) positions in decentralized finance (DeFi) markets, specifically focusing on Constant Product Market Makers (CPMMs). The proposed method constructs optimal mean-variance portfolios that are hedged by short positions in underlying tokens, thereby maximizing the Sharpe ratio. Unlike previous studies, which typically focus on single-pool strategies, our approach integrates multiple CPMM pools and accounts for transaction costs, offering a more comprehensive solution for risk management in DeFi. The model is rigorously tested on historical data from Uniswap V2, demonstrating significant outperformance over traditional liquidity provision strategies, with Sharpe ratios reaching up to 4. Additionally, the framework is shown to be scalable, maintaining positive returns even under high transaction costs and varying capital conditions. This work not only extends existing literature on DeFi liquidity provision but also provides a practical, implementable strategy for diversified portfolio management in volatile cryptocurrency markets.**

***Index Terms*—portfolio management, risk, decentralized exchanges, uniswap, defi**

I. Introduction

As complexity of the decentralized finance (DeFi) landscape increases with introduction of new chains and protocols, the need for rigorous quantitative models and mechanisms for risk diversification is becoming clear. One of the more classic DeFi investment strategies is providing liquidity to a liquidity pool [35] of a *decentralized exchange* [36]. Generally, the liquidity provider remains passive, automatically countertrading every incoming market order, while the price at which trades are executed is determined by an algorithm specified in the exchange's smart contract. This algorithm is called the *Automatic Market Maker (AMM)* [37].

One of the fundamental mechanisms of automatic market makers is the Constant Product Market Maker (CPMM) [1]. In CPMM the price adjustment mechanism is implemented as follows (accrual of trading fees are omitted for simplicity):

1) Balances of tokens x, y provided to the pool by liquidity providers are maintained. The token proportions in the pool must be 1:1 in the equivalent quote.
2) The invariant $D = x \cdot y$ is calculated.
3) When a market order is executed, the invariant D must be preserved, i.e. when Δx is added, the equation $(x + \Delta x) \cdot (y - \Delta y) = D = x \cdot y$ must be solved with respect to Δy.
4) This equation has a simple analytical solution. Moreover, it is easy to show that the instantaneous price of the second token (when $\Delta x \to 0$) is equal to the ratio of tokens in the pool, i.e., $\frac{x}{y}$.

This algorithm serves as a foundation for more complex AMMs, such as Balancer V2 [8], Curve's CryptoSwap [9] and Uniswap V3 [10], while also maintaining a fair share of market. For newly issued tokens, perhaps not observed enough for elaborate use of more sophisticated techniques, CPMM-based pools often have the highest trading activity. The analytical properties of CPMMs are also attractive to researchers due to their relative simplicity, while inferences are often transferable to a broader class of AMMs (the way [2] generalizes to concentrated liquidity positions using results of [3]).

Liquidity providers in non-oracle-based AMMs are exposed to DEX-specific impermanent loss [11], as well as more common types of market risks. They are compensated for the former with *trading fees* - a percentage of any trade (constant [10] or variable [9]) adds to the pool liquidity, which gradually increases its total value. However, expected returns are hard to estimate, and the best way to determine the optimal hedging technique is to take a look at the following.

Optimal liquidity provision for decentralized exchanges is represented in the literature [17], [32]–[34], while the greatest interest among researchers, especially in the case of Constant Function Markets (i.e., maintaining the property $f(x, y) = D$ similar to CPMM, where $f(x, y) = x \cdot y$), lies in the hedging and replication of LP positions, possibly using with derivatives. Replication is generally divided into static, where the replicating portfolio is fixed for the entire lifetime of the position, and dynamic, where rebalancing occurs periodically. Static replication schemes for LP positions are proposed in [2], [3], while dynamic replication is discussed in [5]. However, most works are limited to considering a single pool and, moreover, a single token (and one numerator cryptocurrency, such as an USD-pegged stablecoin).

All of the above emphasizes the importance and relevance of developing a method to build a diversified portfolio of LP positions, optimal from a risk-return perspective. This paper proposes an method for constructing optimal mean-variance portfolios of tokens and CPMM-based liquidity pools, which is also generalizable to other, more complex AMMs and DeFi instruments.

A. Paper's Outline

In Section III a quantitative model for an LP-position is proposed and analytical expressions for covariances are

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derived. The results allow for formulating and solving an optimal portfolio task of mean variation, which is done in Section IV. Section V contains backtesting results and analysis on structure of optimal portfolios.

II. Related Work

To construct optimal portfolios of liquidity positions in Constant Product Market Maker (CPMM) pools, various quantitative models and optimization methods can be employed. One key approach is the Mean-Variance portfolio optimization method [15]. Automated Market Makers (AMMs) play a crucial role in this context by implementing peer-to-peer methods where liquidity providers contribute assets to liquidity pools, facilitating asset exchange within these pools [18]. Liquidity pools managed by AMM smart contracts handle significant token pairs, as seen in Uniswap V2 [19].

Constant Product Market Makers (CPMMs) have been studied extensively, with research focusing on profitability frontiers and lower bounds of assets within liquidity pools [20]. Delta hedging liquidity positions on AMMs like Uniswap v2 and v3 has been addressed through algorithmic approaches [21]. Additionally, strategies for liquidity providers in pools with low price volatilities have been explored, highlighting the trade-offs between risks and returns [22].

Optimal portfolio selection techniques, such as multi-portfolio optimization and factor models for credit risk management, offer insights into combining multiple accounts while considering specific constraints and expected returns [23], [24]. Furthermore, studies on the optimal liquidation of portfolios in dark pools and continuous-time models provide frameworks to manage execution uncertainty and price impact costs [25], [26].

In the context of portfolio optimization, approaches like Cauchy-Schwarz maximization and Nash equilibrium considerations have been proposed to enhance mean-variance portfolio selection [27], [28]. Moreover, the use of meta-optimization techniques and adaptive portfolio combinations from existing algorithms can aid in portfolio construction [29].

CPMMs present unique challenges such as cyclic arbitrage opportunities due to temporary price inaccuracies across liquidity pools [30]. Understanding liquidity provider returns in geometric mean markets and analyzing Uniswap markets can provide further insights into optimizing liquidity positions within CPMM pools [31].

Our work extends these foundations by proposing a comprehensive framework for constructing optimal portfolios of LP positions across multiple CPMM pools, integrating hedging positions and transaction costs. Unlike previous studies, we derive analytical expressions for covariances between LP positions and tokens, incorporate more sophisticated fee accrual modeling, and introduce a novel method for including gas costs in the optimization process. Furthermore, we conduct extensive backtesting across various market conditions and analyze scalability, providing practical insights that were largely absent from theoretical treatments in prior literature. This holistic approach bridges the gap between single-pool theoretical models and the practical challenges of managing diversified liquidity portfolios in DeFi markets, offering a more robust and implementable strategy for liquidity provision.

III. Model for CPMM LP Position Value

Definitions

Let's consider a time period $(T, T+t)$. The price of a cryptocurrency is p_1 at the beginning of the period and p_2 at the end of the period. For integrity of presentation, the following notation is introduced for price changes:

- Pct-returns (over this time period): $\frac{p_2}{p_1}$
- Log-returns: $\log(\frac{p_2}{p_1})$

A. Distribution of CPMM Position Value Under GBM

The geometric Brownian motion is used as a model for the joint distribution of the underlying token prices. Here, log-returns of an individual token for the minimum time interval t, ξ_t, are distributed as $\xi_T \sim \mathcal{N}(\mu_T, \sigma_T^2)$. It is assumed that the increments of the process are independent, thus the model properties do not depend on the discretization frequency.

The volatility and covariances of the log-returns are considered external parameters. At the same time, μ is determined by the no-arbitrage principle, which in its simplest form requires the identity $\mathbb{E}e^{\xi} = 1$. This condition is satisfied by $\mu = -\frac{\sigma^2}{2}$ [12].

Therefore, $(\xi_1, \xi_2) \sim \mathcal{N}\left(\begin{bmatrix} -\frac{\sigma_1^2}{2} \\ -\frac{\sigma_2^2}{2} \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{1,2} & \sigma_1^2 \end{bmatrix}\right)$, where $\sigma_i^2 = A_{i,i}$, $\sigma_{1,2} = A_{1,2}$.

Say in the beginning of a period x units of token 1 and y units of token 2 were added to the liquidity pool. CPMM requires that the proportion of the values of the tokens added to be 1:1. Then the log-return of the LP position is distributed normally, and parameters of the distribution are in a simple relationship with parameters of underlying tokens, as shown in the following statement:

Theorem 1. *Let the log-returns of the two tokens in the CPMM pool be distributed as*

$$(\xi_1, \xi_2) \sim \mathcal{N}\left(\begin{bmatrix} -\frac{\sigma_1^2}{2} \\ -\frac{\sigma_2^2}{2} \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{1,2} & \sigma_1^2 \end{bmatrix}\right)$$

Then, neglecting commissions, the log-return of the LP position is distributed as

$$\mathcal{N}(\mu_{LP_{1,2}}, \sigma_{LP_{1,2}}^2),$$

$$\textit{where } \mu_{LPz_{1,2}} = -\frac{\sigma_1^2 + \sigma_2^2}{4}, \sigma_{LP_{1,2}}^2 = \frac{\sigma_1^2 + \sigma_2^2 + 2\sigma_{1,2}}{4} \quad (1)$$

Proof: Let the token prices at the beginning of the period be p_1 and p_2 respectively. Then, by the CPMM construction, the equality $x \cdot p_1 = y \cdot p_2$ must hold, from which $x = y\frac{p_2}{p_1}$. The invariant of position D can be expressed as

$$D = y^2 \frac{p_2}{p_1} = x^2 \frac{p_1}{p_2} \quad (2)$$

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Then the value of the position at the beginning of the period is

$$\mathrm{LP}_{\mathrm{start}} = x \cdot p_1 + y \cdot p_2 = 2 \cdot y \cdot p_2 = \\ = \sqrt{\frac{D}{\frac{p_2}{p_1}}} \cdot p_2 \cdot 2 = \sqrt{D} \cdot \sqrt{p_1}\sqrt{p_2} \cdot 2 \tag{3}$$

Now consider the end of the period, when the token prices are $p_1 \cdot e^{\xi_1}$ and $p_2 \cdot e^{\xi_2}$ respectively. The invariant D remains unchanged, but the quantity of each type of token in the pool changes; let us denote them as $\hat{x}$ and $\hat{y}$. Similarly to (2):

$$D = \hat{y}^2 \frac{p_2 \cdot e^{\xi_2}}{p_1 \cdot e^{\xi_1}} = \hat{x}^2 \frac{p_1 \cdot e^{\xi_1}}{p_2 \cdot e^{\xi_2}}$$

and the new value of the position will be

$$\begin{aligned} \mathrm{LP}_{\mathrm{end}} &= \sqrt{D} \cdot \sqrt{p_1 \cdot e^{\xi_1}}\sqrt{p_2 \cdot e^{\xi_2}} \cdot 2 \\ &= \sqrt{D} \cdot \sqrt{p_1}\sqrt{p_2} \cdot 2 \cdot \sqrt{e^{\xi_1}}\sqrt{e^{\xi_2}} \\ &= \mathrm{LP}_{\mathrm{start}} \cdot e^{\frac{\xi_1+\xi_2}{2}} \end{aligned} \tag{4}$$

Hence

$$\begin{aligned} \log \frac{\mathrm{LP}_{\mathrm{end}}}{\mathrm{LP}_{\mathrm{start}}} &= \frac{\xi_1 + \xi_2}{2} \\ &\sim \mathcal{N}\left(\frac{\mu_1+\mu_2}{2}, \left(\frac{\sigma_1}{2}\right)^2 + \left(\frac{\sigma_2}{2}\right)^2 + \rho_{1,2}\frac{\sigma_1\sigma_2}{2}\right) \\ &= \mathcal{N}\left(-\frac{\sigma_1^2+\sigma_2^2}{4}, \frac{\sigma_1^2+\sigma_2^2+2\sigma_{1,2}}{4}\right) \end{aligned} \tag{5}$$

■

It is worth noting that the no-arbitrage condition (NAC) $\mu = -\frac{\sigma^2}{2}$ does not generally hold for the LP-position. Instead, the drift term is strictly less then the one required by NAC, unless two tokens are identical; therefore, expected pct-return of the position is always less than 1. This happens due to *impermanent loss*.

B. Covariation matrices

Covariances between instruments are of particular importance in the context of portfolio optimization. In this chapter calculations for covariances between cryptocurrencies and liquidity positions are provided, starting from log-returns and following with pct-returns.

We start with two trivial statements on covariations of log-returns.

Statement 1. *Consider the case of a token and an LP position. Let i, j, k be the indices of the tokens, and the liquidity pool consists of tokens j, k. Then the covariance of the log-returns of the LP position and token i is:*

$$\begin{aligned} \mathrm{Cov}(\xi_i, \frac{\xi_j+\xi_k}{2}) &= \frac{1}{2}(\mathrm{Cov}(\xi_i,\xi_j) + \mathrm{Cov}(\xi_i,\xi_k)) \\ &= \frac{\sigma_{i,j}+\sigma_{i,k}}{2} \end{aligned} \tag{6}$$

Statement 2. *Consider the case of two LP positions. Let i, j, k, l be the indices of the tokens, the first liquidity pool consists of tokens i, j, and the second of k, l. Then, similarly, the covariance of the log-returns of the LP positions is:*

$$\mathrm{Cov}(\frac{\xi_i+\xi_j}{2}, \frac{\xi_k+\xi_l}{2}) = \frac{\sigma_{i,k}+\sigma_{i,l}+\sigma_{j,k}+\sigma_{j,l}}{4} \tag{7}$$

Calculating the portfolio covariance, on the other hand, requires expressions for covariances of pct-returns.

Lemma 1. *Let*

$$(\xi_1, \xi_2) \sim \mathcal{N}\left(\begin{bmatrix} -\frac{\sigma_1^2}{2} \\ -\frac{\sigma_2^2}{2} \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{1,2} & \sigma_2^2 \end{bmatrix}\right)$$

Then

$$Cov(e^{\xi_1}, e^{\xi_2}) = \exp(\mu_1+\mu_2+\frac{\sigma_1^2+\sigma_2^2}{2})(\exp(\sigma_{1,2})-1) \tag{8}$$

Proof:

By definition, $\mathrm{Cov}(e^{\xi_1}, e^{\xi_2}) = \mathbb{E}(e^{\xi_1}e^{\xi_2}) - \mathbb{E}(e^{\xi_1})\mathbb{E}(e^{\xi_2})$

Let's obtain the expressions for $\mathbb{E}(e^{\xi_1})$, $\mathbb{E}(e^{\xi_2})$, and $\mathbb{E}(e^{\xi_1+\xi_2})$ separately.

1) $\forall i\ \mathbb{E}(e^{\xi_i}) = e^{\mu_i + \frac{\sigma_i^2}{2}}$
2) $\mathbb{E}(e^{\xi_1+\xi_2}) = e^{\mu_1+\mu_2+\frac{\sigma_1^2+\sigma_2^2+2\sigma_{1,2}}{2}}$

Now substitute these values into the formula for the covariance:

$$\begin{aligned} \mathrm{Cov}(e^{\xi_1}, e^{\xi_2}) &= \exp(\mu_1+\mu_2+\frac{\sigma_1^2+\sigma_2^2+2\sigma_{1,2}}{2}) \\ &\quad - \exp(\mu_1+\frac{\sigma_1^2}{2}) \cdot \exp(\mu_2+\frac{\sigma_2^2}{2}) \\ &= \exp(\mu_1+\mu_2+\frac{\sigma_1^2+\sigma_2^2}{2}) \cdot \exp(\sigma_{1,2}) \\ &\quad - \exp(\mu_1+\mu_2+\frac{\sigma_1^2+\sigma_2^2}{2}) \\ &= \exp(\mu_1+\mu_2+\frac{\sigma_1^2+\sigma_2^2}{2}) \cdot (\exp(\sigma_{1,2})-1) \end{aligned} \tag{9}$$

■

Using this lemma, it is straightforward to derive covariances for pct-returns in pool-crypto and pool-pool pairs. The exact formulas are:

$$\begin{aligned} \mathrm{Cov}(e^{\xi_i}, \mathrm{LP}_{j,k}) = &\exp(-\frac{\sigma_j^2+\sigma_k^2}{8} + \frac{\sigma_{j,k}}{4}) \\ &\cdot (\exp(\frac{\sigma_{i,j}+\sigma_{i,k}}{2}) - 1) \end{aligned} \tag{10}$$

$$\begin{aligned} \mathrm{Cov}(\mathrm{LP}_{i,j}, \mathrm{LP}_{k,l}) = &\exp\left(-\frac{\sigma_i^2+\sigma_j^2+\sigma_k^2+\sigma_l^2}{8}\right) \\ &\cdot \exp\left(\frac{\sigma_{i,j}+\sigma_{k,l}}{4}\right) \\ &\cdot \left(\exp\left(\frac{\sigma_{i,k}+\sigma_{i,l}+\sigma_{j,k}+\sigma_{j,l}}{4}\right) - 1\right) \end{aligned} \tag{11}$$

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C. Expected value of liquidity position

We also need a formula for the expected pct-return of the LP position for calculations. Unlike the component tokens, the identity $\mathbb{E}e^{\xi} = 1$ does not generally hold for the pool's return. Instead, for the case without trading fees:

$$\begin{aligned}\mathbb{E}e^{\xi_{\mathrm{LP}_{i,j}}} &= \exp(\mu_{\mathrm{LP}_{i,j}} + \frac{\sigma^2_{\mathrm{LP}_{i,j}}}{2}) \\ &\overset{(1)}{=} \exp(-\frac{\sigma_i^2+\sigma_j^2}{8} + \frac{\sigma_{i,j}}{4}) \\ &= \exp(-\frac{\sigma_i^2+\sigma_j^2-2\sigma_i\sigma_j\rho_{i,j}}{8}) \end{aligned} \tag{12}$$

Additionally, in the formula for the expected value of the LP position, it is necessary to account for the yield from commissions. In this work it is considered as a constant parameter for each pool and investment period, without any stochasticity. Let APY be the expected annual percentage yield from commissions, accounting for compound interest. APY is assumed to be constant in the absence of fund inflows and outflows from the pool. Then

$$\mathbb{E}e^{\xi_{\mathrm{LP}_{fee|i,j}}} \overset{(12)}{=} \exp(-\frac{\sigma_i^2+\sigma_j^2-2\sigma_i\sigma_j\rho_{i,j}}{8}) \cdot (1+\phi)$$

$$\text{where } \phi = (1+\mathrm{APY})^{\tau} - 1$$

Here, ϕ is the expected return from fees over the period τ.

Note that this formula considers the return of a position already deposited in the pool, where the fee yield is calculated assuming the pool's size does not change. In realistic scenarios it is necessary to estimate the fee yield ϕ_{strat} considering that additional funds will be deposited into the pool, which, in the case of small pools, may be commensurate with the total liquidity value (TVL).

Let the expected volume over the period τ be V_τ, the commission rate be ν, and the current pool size be TVL. Then

$$\phi = \frac{V_\tau \cdot \nu}{\mathrm{TVL}}$$

Let A dollars of liquidity be additionally deposited into the pool. Then our proposed estimate for ϕ_{strat} is expressed as:

$$\phi_{\mathrm{strat}} = \frac{V_\tau \cdot \nu}{\mathrm{TVL}+A} = \phi \cdot \frac{\mathrm{TVL}}{\mathrm{TVL}+A} = \frac{\phi}{1+\frac{A}{\mathrm{TVL}}} \tag{13}$$

The τ for practical applications would be the rebalancing frequency.

Thus, the expected pct-return of the position, considering the new deposit, is equal to

$$\mathbb{E}e^{\xi_{\mathrm{LP}_{\mathrm{strat}|i,j}}} = \exp(-\frac{\sigma_i^2+\sigma_j^2-2\sigma_i\sigma_j\rho_{i,j}}{8}) \cdot (1+\frac{\phi}{1+\frac{A}{\mathrm{TVL}}}) \tag{14}$$

D. Criterion of profitability

The previous conclusions allow us to estimate the required expected return for liquidity provision to be economically viable. From (12), we can derive the necessary APY value to satisfy the inequality $\mathbb{E}e^{\xi_{\mathrm{LP}_{i,j}}} \geq 1$:

$$1+\phi = (1+\mathrm{APY})^{\tau} \geq \exp(\frac{(\sigma_i-\sigma_j)^2+2\sigma_i\sigma_j(1-\rho_{i,j})}{8}) \Leftrightarrow$$

$$\Leftrightarrow \mathrm{APY} \geq \exp(\frac{(\sigma_i-\sigma_j)^2+2\sigma_i\sigma_j(1-\rho_{i,j})}{8\tau}) - 1$$

The parameters $\sigma_{i,j}$ in the formula also denote volatility over the period τ, and thus are scaled by $\sqrt{\tau}$. Transitioning to annualized volatility $\hat{\sigma}_{i,j}$, we obtain

$$\mathrm{APY} \geq \exp(\frac{(\hat{\sigma}_i-\hat{\sigma}_j)^2+2\hat{\sigma}_i\hat{\sigma}_j(1-\rho_{i,j})}{8}) - 1 \tag{15}$$

IV. Optimal Portfolios

In this section, we propose a procedure for finding optimal portfolios of liquidity positions and tokens, the process outline in Figure 1.

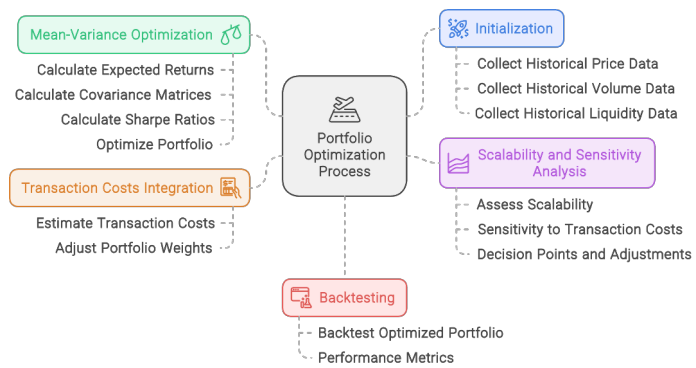

Fig. 1: Optimal Liquidity Position Portfolios in Decentralized Finance.

We associate portfolios with allocation vectors w_i, considering both liquidity positions and token positions as instruments. The target function for maximization is the Sharpe ratio, with zero risk free rate:

$$\mathrm{Sharpe}_\tau = \frac{\mathbb{E}(r_\tau)}{\sigma(r_\tau)} \tag{16}$$

Let the number of pools be P, then instruments $1..P$ are pools, and $P+1..N$ are tokens. Let each instrument be allocated a weight w_i. Then the following constraints are imposed:

1) $\forall i \in \{1..P\}\ w_i \geq 0$
2) $\forall i \in \{P+1..N\}\ w_i \leq 0$
3) $\sum_{i=1}^{N} |w_i| = 1$

i.e. all positions in liquidity pools are *non-negative*, and all positions in tokens are *non-positive*. The latter serves as a form of regularization, as positions in tokens should hedge market risk.

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The denominator of the Sharpe ratio is calculated as

$$\sigma(r) = \sqrt{\mathbb{D}r} = \sqrt{\sum_{i=1}^{N}\sum_{j=1}^{N} w_i \cdot w_j \cdot \mathrm{Cov}_{pct}(i,j)}$$

where the covariances $\mathrm{Cov}_{pct}(i,j)$ use equations (10) and (11).

Let r_i be the expected pct-return (minus one) for instrument i. Then

$$\mathbb{E}(r) = \sum_{i=1}^{N} w_i \cdot r_i$$

The expected return r_i is calculated as follows:

1) For tokens, the expected return is zero:

$$\forall i \in \{P+1..N\}\ r_i = 0$$

2) For pools, the return is calculated using formula (14) minus one:

$$\forall i \in \{1..P\}\ r_i = \exp\left(-\frac{\sigma_1^2 + \sigma_2^2 - 2\sigma_1\sigma_2\rho_{1,2}}{8}\right) \cdot \left(1 + \frac{\phi_i}{1 + \frac{w_i \cdot \mathrm{AUM}}{\mathrm{TVL}_i}}\right) - 1 \qquad (17)$$

Here, ϕ_i is the expected pool return from commissions (as pct-return minus one, for the period τ), AUM is the total deposit in the strategy, and TVL_i is the total value of liquidity in pool i.

Due to taking the deposit size into account, the optimization problem $\mathrm{Sharpe}_\tau \xrightarrow{w_{1..N}} \max$ becomes a non-linear programming problem.

The proposed optimization procedure consists of the following steps:

1) Exclude pools that do not satisfy the profitability condition (15), and then exclude instruments that are not present in any pool. This step significantly reduces the dimensionality of the optimization problem.
2) Calculate the variance-covariance matrix for pools and tokens.
3) Compute the vector $\phi_{\text{init}} = \left(\frac{\phi_i}{1+\frac{\mathrm{AUM}}{\mathrm{TVL}_i}}\right)\big|_{i=1..P}$.
4) For all pools, set proportional allocations $w_{\text{init}|p} \sim \phi_{\text{init}|p},\ p \in \{1..P\}$. For a token $i,\ i \in \{P+1..N\}$ that is in pool p, set the weight $w_{\text{init}|i} = -\frac{w_{\text{init}|p}}{2}$. If the token is a stablecoin, it is not assigned a weight. The vector w_{init} serves as the initial guess for the optimizer.
5) Optimize the Sharpe ratio (16) with $r_0 = 0$ using the Limited-memory BFGS method [14] (and its implementation `L-BFGS-B` in the `scipy.optimize` package) with sign constraints on the allocations, adding $\varepsilon = 10^{-5}$ to the denominator to avoid numerical errors. The maximum number of iterations is 1000.
6) Normalize the final weight vector to one to satisfy the total allocation condition.

A. Optimization with transaction costs

To optimize for Sharpe ratio considering transaction costs (i.e. gas cost), a continuous approximation of the loss function is used. If the cost of a single transaction (deposit or withdrawal from a liquidity pool) is G in quote currency, the corresponding term is

$$\mathrm{Cost} = \sum_{i=1}^{P} l(|w_i - v_i|) \cdot G, \text{ where } l(d) = \min(|d| \cdot 10, |d|^{\frac{1}{5}})$$

It is chosen to have all the desirable properties of a barrier function [13]: continuity and bound derivative (that exists almost everywhere), while also being a good approximation to an indicator function $I\{|d| > 0\}$.

Now, let r be the portfolio return calculated by the base procedure. Then the final gas-adjusted returns return is

$$r_{\text{gas}} = (1+r) \cdot \left(1 - \frac{\mathrm{Cost}}{\mathrm{AUM}}\right)$$

This expression becomes the numerator when calculating the objective function (the Sharpe ratio).

V. Backtesting and Results

The dataset consists of data on Uniswap V2 pools on the Ethereum Mainnet. The period under consideration is approximately 3 years: 2021-01-13 to 2024-01-01. Due to the permissionless nature of the protocol, it is not feasible to consider all pools that existed during this period, so a filtering procedure was implemented.

Data selection for consideration was carried out as follows.

1) Market data with daily granularity for the specified period was obtained for the Top-1000 cryptocurrencies from the CoinGecko website.
2) The following were excluded:
   - Tokens for which data (market capitalization, volume, price) had missing values after their first appearance
   - Stablecoins (replaced with a constant rate of 1 USD)
3) Wrapped Bitcoin and Wrapped Ethereum were added.
4) Data on the Top-100 pools by trading volume for each day was obtained through the Uniswap V2 Subgraph [7].
5) For any pool that appeared in the Top-100, data for the following 4 weeks was obtained (if this data was not already in the sample).
6) Pools consisting of two stablecoins, pools for which market data for at least one token is unavailable, and pools with liquidity less than \$50,000 were excluded.

Future fee accrual for optimization was estimated using historical data as average of fee-induced log-returns for the position over last week, with an upper bound of 0.3% · TVL per day (which corresponds to daily volume equal to total pool TVL). The actual fee accrual estimate for the calculation of PnL was done with daily data corrected for the deposit size, similar to (13). The covariation matrix was estimated using daily returns for the last 25 weeks and normalized to a weekly

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scale; tokens with a market data history of less than 25 weeks were excluded.

The strategies tested used weekly rebalancing. Annualized Sharpe ratio with risk-free ratio r_0 of 2%, Maximal Drawdown (MD) and Annualized Percentage Yield (APY) were used as strategy metrics. All strategies were limited to 20 instruments (*quantized*) for purposes of robustness, unless stated otherwise.

A. Strategy performance

In this section, a comparison of the baseline strategy with benchmarks is performed in different stages of the market. The following strategies are chosen as benchmarks:

1) *Top-pool-autohedge (TPA)* : At each step, the pool with the highest expected return is selected. Half of the deposit is allocated to this pool; the remaining half is used for hedging positions corresponding to the pool's composition.
2) *Uniform-autohedge (UA)*: Funds are evenly distributed across all available pools and hedged in the same manner.
3) *Top-500-crypto (TOP500)*: A market-cap weighted index based on the top 500 tokens.

Proposed strategy is marked as "Our".

Both proposed stratedgy and benchmark strategies *TPA* and *UA* assume a fund volume of $100,000. Quantization is not used on the benchmarks.

The market stages selected correspond to the years 2021, 2022, and 2023. These periods were chosen because the market sentiment changed significantly, making them suitable for investigating the robustness of the proposed strategies to trending movements. The results are presented in Table I.

TABLE I: Comparison with benchmarks

Period	**Strategy**	Sharpe	Max Drawdown	APY
2021	*TPA*	**1.78**	4.40%	16.71%
	UA	-1.54	8.62%	-7.78%
	TOP500	1.52	50.83%	**246.33%**
	Our	1.66	**3.67%**	10.62%
2022	*TPA*	1.69	4.97%	12.70%
	UA	4.20	**0.16%**	4.03%
	TOP500	-1.47	64.00%	-62.41%
	Our	**6.01**	1.67%	**19.09%**
2023	*TPA*	2.93	2.58%	15.49%
	UA	0.54	**0.68%**	2.47%
	TOP500	2.17	19.48%	**149.47%**
	Our	**6.75**	1.00%	19.24%
2021-2023	*TPA*	1.98	4.97%	15.01%
	UA	-0.62	8.62%	-0.37%
	TOP500	0.52	72.35%	**44.21%**
	Our	**3.82**	**3.67%**	16.46%

The proposed strategy outperforms benchmarks by a large margin in risk-adjusted return metrics when considering the whole testing period and demonstrates an outstanding risk-return ratio even during a negative trend when the market index dropped by 62.4% (in 2022).

B. Scalability

Table II shows response of the strategies to increase in capital under management. As expected, Sharpe ratio declines with capital growth and earning capacity of high-activity pools becoming insufficient. However, Sharpe ratios remain positive for AUM up to $10M, and all strategies are characterized by low risk in terms of Maximum Drawdown.

TABLE II: Strategy scalability

AUM (USD)	Sharpe	MD	APY
≤ 1	4.41	4.885%	**21.40%**
1,000	4.26	3.614%	20.17%
10,000	**4.91**	3.457%	21.21%
100,000	3.82	3.669%	16.46%
1,000,000	1.65	3.902%	7.82%
10,000,000	0.76	**2.762%**	4.37%

It is worth noting that by construction, the portfolio must deploy all funds either in LP positions or in hedging positions in tokens, which explains the observed decline in the Sharpe ratio: without this condition and with the possibility of allocating funds at a risk-free rate, the portfolio optimal in terms of the Sharpe ratio would be one with the minimum amount of funds actually invested in the strategy (and proportionally reduced returns). Therefore, a leverage-agnostic metric such as Sharpe ratio may not be optimal for managing large portfolios in real scenarios. One possible solution for this is optimisation for portfolio risk subject to a profitability constraint, with total deployed funds as an additional optimization parameter.

C. Transaction costs

Let the cost of one transaction be denoted as G. The impact of different values of G on the risk and return of the strategies is shown in Table III. On the right are the data for the strategies formed considering transaction costs according to the procedure described in section IV-A.

TABLE III: Performance considering transaction costs

G, USD	**Baseline Strategy**			**Optimization w/ G**		
	Sharpe	MD	APY	Sharpe	MD	APY
1	3.84	**3.70%**	16.50%	**4.08**	3.93%	**17.24%**
5	3.30	**3.84%**	14.39%	**3.33**	4.13%	**15.03%**
20	1.29	**5.68%**	6.78%	**1.42**	5.90%	**7.72%**

Increasing the transaction cost has a significant negative impact on the profitability of the strategy; however, it remains profitable at all considered values. Considering transaction costs when constructing the portfolio *increases* the Sharpe ratio and annual yield, but the maximum drawdown also increases.

D. Structure of optimal portfolios

The conditions on hedging token positions in the optimisation problem are not strict; the token set, proportions and total value of the hedging positions could be arbitrary, as long as every position is short. However, in practice the

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hedging positions in optimal portfolios found by the numerical procedure closely resemble those constructed manually in the initial vector or benchmarks: for a deposit of w in a pool with two tokens each token is assigned weights $-\frac{w}{2}$ (except for stablecoins). The mean weighted difference between those allocations is found to be 0.77% for the $100K strategy and of comparable scale for others. This might suggest further decrease of optimisation problem dimensionality is possible without decreases in the output quality.

VI. Scope and Limitations

Several assumptions and limitations of this work must be addressed to ensure correct interpretation of the results.

A. Model assumptions

First, the quantitative model assumes log-normality of returns of the assets. It is an established industry practice to model cryptocurrency returns using Brownian motion or its local volatility modifications [34], [38], [39]. However, [40], [41] show that the empirical return distribution shape is not normal and is, indeed, high-peaked and thick-tailed. Modelling cryptocurrency returns with a technique using a novel Asymmetric Laplace distribution family has been proposed to account for those findings [42]. The effect of assumption of return normality on this work is, however, limited: normal distributions are explicitly used in calculation of covariation between assets and pools, while the portfolio is constructed based on the general Mean-Variance Portfolio Optimization framework. Empirical deviations from the given assumptions therefore would lead to decrease in accuracy of the covariance approximation. Market-neutrality of the resulting strategy serves as protection from sharp general market downturns or upturns.

An assumption of an infinitely liquid market allowing for both long and short positions in tokens is also made. While there is no such market (to date), instruments like lending protocols allow for similar properties, such as no fixed share fee on borrowing, which is a proxy for a short position in the context of proposed strategies. Lending protocols, however, impose borrowing rates. While supplying a stablecoin as lendable collateral and borrowing a risky asset generally results in a net positive rate (for the lender) due to established supply and demand on lending markets, this is not guaranteed. Both AMM- and CLOB-based markets experience price impact when large orders are executed, as well as exchange fees. Decentralized perpetual futures exchanges also implement funding fees, which can be both favorable and unfavorable for a specific (long / short) position type, leading to introduction of another risk factor. Overall, liquidity constraints and trading fees in hedging markets can significantly impact the strategy and should be accounted for in each specific case.

B. Backtest construction

The backtesting implemented in the paper models transaction fees as constant values, equal for deposit and withdrawal of liquidity. In reality, transaction fees are not constant over time and vary due to different network activity, which may pose a significant challenge when rebalancing a strategy. In real investment scenarios unusually high transaction fees may be countered with use of tokenised gas, changes in timing and use of blockchain networks with lower transaction fees, such as Ethereum L2 networks. The proposed portfolio optimization procedure allows for optimization using current gas price for each moment before anticipated rebalancing.

There is a degree of data selection bias introduced during gathering data for backtesting. Data includes tokens without missing market data and present in Top-1000 at the time of collection. These characteristics would not be known at the time of testing. However, the strategy proposed in this work hedges directional risk, which addresses the possible concern of almost exclusive exposure to long-term gainer cryptocurrencies. Absense of a token in the Top-1000 or missing values in data could potentially indicate insreased *divergence* risk, which significantly affects the strategy performance, but evidence of that is yet to be found.

VII. Conclusion

A. Results and discussion

The findings of this paper propose a framework for constructing portfolios of LP positions hedged by underlying tokens, allowing liquidity providers to earn fees while diversifying their exposure to market risk and impermanent loss. A strategy implementing weekly rebalancing was tested on historical data and shows great ratios of return to risk, surpassing the market index and benchmark strategies. It also is robust to introduction of transaction costs, although performance declines significantly with increase in cost of a single transaction.

We see the main contribution of this paper not in providing a ready-for-trading strategy, but in proposing a framework that can be generalized to other instruments, as well as extending the existing literature on optimal liquidity provision to a multiple-pool scenario.

B. Further research

The proposed framework is generalizable to other DeFi products. It is pretty straightforward to introduce lending protocols into the framework; at least in assumption of known future interest rates. More complex scenario arises when considering concentrated liquidity AMMs, such as Uniswap V3, or liquidity provision to oracle-enabled exchanges. These may require numerical methods and sophisticated heuristics to calculate covariances and expected returns. There is also a connection between volatility and trading volumes (and therefore income from trading fees), which can be unified in a single asset pricing / risk explanation model, such as IPCA [6].

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